

The MOND Acceleration Scale as the Cosmological Constant: a Coefficient-Free Bridge Between Galaxy Dynamics and Dark-Energy Evolution

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Abstract

The MOND acceleration scale a_0 and the cosmological constant Λ are numerically related by $a_0 \approx c^2 \sqrt{\Lambda/32\pi}$, a coincidence noted in various forms by Milgrom, Blanchet, and others. We take this relation literally— a_0 is the cosmological constant expressed as a galactic acceleration—and draw out its sharpest consequence. Read as a relation to the dark energy (not the total density), the relation predicts that the galactically-measured $a_0(z)$ and the cosmologically-measured $\sqrt{\rho_{\text{DE}}(z)}$ are the *same curve*: $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)}$, a one-to-one bridge between two independent measurements in which the (route-forced, undetermined) 32π coefficient *cancels*. This decouples the framework’s empirical test from its hardest theoretical debt. We exhibit a concrete Aether-Scalar-Tensor (AeST) realization that carries $a_0 \propto \sqrt{\rho_{\text{DE}}}$ and verify it is ghost-free, has $c_{\text{GW}} = c$, is CMB-safe at linear order, and keeps a_0 universal across galaxies. We specify the decisive measurement—a ~ 30 – 80 -galaxy JWST+ALMA campaign at $z \sim 3$ —and state the framework’s honest limits: the coefficient is traceable but not uniquely derived, the relation reproduces the value of a_0 only to $\sim 30\%$, the covariant coupling is inserted rather than forced, the distinctive (a_0 -declining) branch is contingent on DESI’s dynamical-dark-energy preference, and the framework is a unification of the dark sector, *not* a theory of everything. Under a constant Λ the prediction is constant a_0 (well-supported MOND with its value explained); under DESI’s evolving dark energy it is a falsifiable, mild decline ($a_0 \approx 0.74 a_0(0)$ at $z = 3$) testable now in principle and decisively at $z \sim 3$.

1 Introduction

The dark sector is $\sim 95\%$ of the cosmic energy budget: $\sim 68\%$ dark energy and $\sim 27\%$ “dark matter.” Modified Newtonian Dynamics [1, 2] replaces the galactic dark-matter *phenomenology* with a modification of gravity below a universal acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, reproducing rotation curves and the radial-acceleration relation (RAR) [3] from the baryons alone with a single constant. It has long been observed [4, 5] that this constant is numerically tied to the cosmological constant,

$$a_0 \approx c^2 \sqrt{\frac{\Lambda}{32\pi}} = \frac{c}{2} \sqrt{G \rho_{\text{DE}}}, \quad (1)$$

with $\rho_{\text{DE}} = \Lambda c^2/8\pi G$ the dark-energy density. Equation (1) reproduces the observed a_0 to $\sim 30\%$. The relation has genuine precedents: Milgrom’s “MOND as a vacuum effect” [4] and $a_0 \sim c\sqrt{\Lambda/3}$

coincidence; Blanchet’s dipolar dark fluid [6], in which Λ is a potential floor and a_0 an anharmonic coefficient; the Cohen–Kaplan–Nelson UV/IR bound [7], which yields $\rho_{\text{DE}} \sim M_P^2 H^2$ of the observed order; and, most recently and most completely, Singh’s relativistic MOND [9], in which the single scale is fixed by a dynamically selected de Sitter radius. The deep-MOND limit itself possesses an exact $SO(4, 1)$ (de Sitter) conformal symmetry in three spatial dimensions [5]; a_0 is the scale at which that symmetry is *explicitly* broken.

This note does not claim to derive (1); the 32π coefficient is traceable to General Relativity (the Einstein coupling 8π and the Friedmann 3) but is route-dependent at the level of a convention, and no mechanism forces it (Sec. 7). Our contribution is to read (1) as a statement about the *dark energy*, derive the coefficient-free bridge it implies (Sec. 3), exhibit a concrete covariant theory that carries it through the CMB (Sec. 4), and specify the measurement that decides it (Sec. 6).

2 The two dark components as one scale

Taken literally, (1) unifies the dark sector: the $\sim 68\%$ dark energy *is* Λ , and the galactic dark-matter phenomenology is the a_0 -modification of gravity sourced by the *same* Λ . Dark matter and dark energy become two faces of the vacuum—95% of the energy budget from a single scale. This is a unification of the dark sector only: the Standard Model (the remaining $\sim 5\%$) is not addressed and is not derivable from the relation. We return to this honestly in Sec. 7.

3 The coefficient-free bridge

A subtlety resolves a long-standing ambiguity. The relation has two forms that agree today but diverge in the past:

$$\underbrace{a_0 = c^2 \sqrt{\Lambda/32\pi} = \frac{c}{2} \sqrt{G\rho_{\text{DE}}}}_{\propto \sqrt{\rho_{\text{DE}}}} \quad \text{vs.} \quad \underbrace{a_0 = cH(z)/Z, \quad H^2 = \frac{8\pi G}{3}\rho_{\text{tot}}, \quad Z = 2\sqrt{8\pi/3}}_{\propto \sqrt{\rho_{\text{tot}}}}. \quad (2)$$

They coincide at $z = 0$ via the “why-now” coincidence $cH_0 \approx c^2\sqrt{\Lambda}$, but ρ_{tot} is matter-dominated in the past (it rises as $(1+z)^3$) while ρ_{DE} is not. The form faithful to “ a_0 is set by the cosmological constant” is $a_0 \propto \sqrt{\rho_{\text{DE}}}$; the $\sqrt{\rho_{\text{tot}}}$ form is the apparent-horizon packaging, and it is disfavoured by existing Tully–Fisher data [11, 12] (which test exactly these two couplings and marginally prefer $\sqrt{\rho_{\text{DE}}}$).

Adopting $a_0 \propto \sqrt{\rho_{\text{DE}}}$, the ratio is independent of the undetermined coefficient:

$$\boxed{\frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE}}(0)}}} \quad (3)$$

since $(c/2)\sqrt{G}$ and 32π cancel. Equation (3) is a one-to-one link between the *galactically*-measured evolution of a_0 and the *cosmologically*-measured dark-energy density. It is falsifiable without ever determining 32π :

cosmology says	galaxies say	verdict
Λ constant ($w = -1$)	a_0 constant	consistent (the safe core)
Λ constant ($w = -1$)	a_0 evolves	falsified
dynamical DE (DESI)	$a_0 \propto \sqrt{\rho_{\text{DE}}}$	confirmed (distinctive)
dynamical DE (DESI)	a_0 constant	falsified

With the DESI DR2 [14] CPL parameters $(w_0, w_a) = (-0.752, -0.86)$, (3) predicts $a_0(z)/a_0(0) = \{1.01, 0.86, 0.74\}$ at $z = \{1, 2, 3\}$ —a mild *decline*, the only reading below unity, and consistent with the high- z Tully–Fisher data that exclude the steep $cH(z)$ rise.

4 A covariant realization

Aether-Scalar-Tensor gravity (AeST) [10]—a metric, a unit-timelike vector A^μ , and a scalar ϕ , governed by a free function $\mathcal{F}(\mathcal{Q}, \mathcal{Y})$ of the cosmological invariant $\mathcal{Q} \sim A^\mu \partial_\mu \phi$ and the spatial-gradient invariant $\mathcal{Y}$ —reproduces MOND and fits the CMB. We realize $a_0 \propto \sqrt{\rho_{\text{DE}}}$ with the non-separable free function

$$\mathcal{F}(\mathcal{Q}, \mathcal{Y}) = K(\mathcal{Q}) + C(\mathcal{Q}) \mathcal{Y}^{3/2} + (\text{Newtonian crossover}), \quad C(\mathcal{Q}) = \kappa_0 \sqrt{\rho_0 / \rho_{\text{DE}}(\mathcal{Q})}. \quad (4)$$

Because the deep-MOND scale obeys $a_0 \propto 1/(\text{the } \mathcal{Y}^{3/2} \text{ coefficient})$, this gives $a_0(\mathcal{Q}) = a_0(0) \sqrt{\rho_{\text{DE}}(\mathcal{Q}) / \rho_0}$ exactly: the bridge (3) falls out of the action. Four checks pass (verified symbolically; see the accompanying repository):

- **Ghost-free / hyperbolic:** $\mathcal{F}_{\mathcal{Y}} = \frac{3}{2} C \sqrt{\mathcal{Y}}$ and $2\mathcal{Y} \mathcal{F}_{\mathcal{Y}\mathcal{Y}} + \mathcal{F}_{\mathcal{Y}} = 3C \sqrt{\mathcal{Y}}$ are both > 0 iff $C > 0$, which holds wherever $\rho_{\text{DE}} > 0$.
- $c_{\text{GW}} = c$: $\mathcal{F}$ is a scalar potential; it does not enter the graviton kinetic term (consistent with GW170817).
- **CMB-safe:** $\mathcal{Y}$ is gradient-built, so $\bar{\mathcal{Y}} = 0$ on FRW and $C(\mathcal{Q}) \mathcal{Y}^{3/2} = \mathcal{O}(\delta^3)$; a_0 is absent from the linear equations regardless of its value, and tying C to the *background* $\mathcal{Q}$ does not promote it.
- **Universality:** $\mathcal{Q}$ is the cosmological time-sector invariant; galaxies source the spatial $\mathcal{Y}$ (the MOND potential) and perturb $\mathcal{Q}$ only at the potential level $\Phi = (V/c)^2 \sim 10^{-6}$, so $\delta a_0 / a_0 \sim 10^{-6} - 10^{-3} \ll$ the RAR scatter ($\sim 29\%$). (Order-of-magnitude; a full quasi-static solution remains.)

The $C(\mathcal{Q}) \rightarrow \infty$ strong-coupling limit (dynamical DE, $\rho_{\text{DE}} \rightarrow 0$ in the deep past) is benign: it occurs at $z \sim 1100$ where no nonlinear structures exist ($\mathcal{Y} \approx 0$), while the first galaxies form at $z \lesssim 10\text{--}20$ where C is only $\sim 3\text{--}5\times$; the two epochs do not overlap, $a_0 \rightarrow 0$ smoothly, and it is a continuation of deep-MOND’s intrinsic $\mathcal{Y}^{3/2}$ strong coupling shared by all MOND theories.

5 Predictions across five independent channels

Because a_0 enters every system below the acceleration scale, the bridge (3) is one coefficient-free amplitude shared across many observables. Parametrizing $a_0(z) = a_0(0) [\rho_{\text{DE}}(z) / \rho_{\text{DE}}(0)]^{\beta/2}$, the framework is $\beta = 1$ and ordinary (constant- a_0) MOND is $\beta = 0$. We computed five channels (all reproducible in the repository); each measures β , the low- z channels anchoring $a_0(0)$ while the high- z channels carry the evolution:

channel	$z = 0$ status	distinctive ($\beta=1$) prediction
mass-discrepancy-acceleration	SPARC, 0.20 dex	deep-MOND discrepancy -12% at $z=3$
dwarf-spheroidal dispersions	5/8 to $\sim 40\%$, EFE	fixed-mass dwarf $\sim 7\%$ colder at $z=3$
galaxy-galaxy lensing	KiDS RAR [15]	saturated $\alpha_\infty = 2\pi \sqrt{GMa_0}/c^2$; non-monotonic
high- z Tully–Fisher	on $z=0$ BTFR (errors)	discs 11–16% <i>below</i> BTFR at $z=4\text{--}6$ (sign test)
external-field effect	SPARC [16], 4–5 σ	EFE strength $\propto 1/\sqrt{\rho_{\text{DE}}}$, +36% by $z=3$

Two features make the set decisive rather than merely consistent. First, ρ_{DE} is nearly flat below $z \sim 2$ (it peaks $\sim 6\%$ near $z \sim 0.4$), so *all* evolution-constraining power lies at $z > 2$; a Fisher combination gives 3σ (5σ) on β from ~ 30 discs at $z = 3$ (~ 25 at $z = 5$ plus a $z \sim 3$ anchor), with redshift outweighing sample size. Second, the external-field effect violates the strong equivalence principle, which a dark-matter halo cannot do; *any* detected EFE evolution is therefore impossible to reproduce with halos, making it the cleanest discriminator. The high- z Tully–Fisher *sign* (discs below, not above, the local relation) already excludes the rising $\sqrt{\rho_{\text{tot}}}$ branch.

6 The decisive measurement

The bridge (3) is tested via the baryonic Tully–Fisher zero point: $M_b = V_{\text{flat}}^4/(Ga_0)$, so at fixed M_b , $V_{\text{flat}} \propto a_0^{1/4}$. The coefficient-free ratio $a_0(z)/a_0(0)$ is read from the shift of the BTFR zero point between a $z \sim 3$ sample and a $z \sim 0$ anchor analyzed through an identical pipeline (so the IMF, stellar-mass, interpolation-function, and absolute- a_0 systematics cancel). The faithful-vs-constant signal at $z = 3$ is a 0.033 dex shift in V_{flat} ; the excluded $cH(z)$ rise (+0.16 dex) is trivially separated. A forecast with per-galaxy precision ~ 0.06 dex gives a 3σ (5σ) separation with ~ 30 (~ 80) low-surface-density, rotation-supported deep-MOND discs at $z = 2.5\text{--}3.5$, using JWST/NIRSpec outer rotation curves, ALMA [C II]/CO gas masses, and JWST SED stellar masses. The measurement requires no new facility and no determination of the coefficient. (DESI itself supplies the $\rho_{\text{DE}}(z)$ input but, being a redshift survey, cannot measure a_0 ’s evolution; its peculiar-velocity component is $z < 0.15$.)

7 Honest assessment

We state the limits plainly, since they bound what (1)–(3) can claim.

1. **Not novel at the core.** The a_0 – Λ relation and the deep-MOND $SO(4, 1)$ /de Sitter structure are Milgrom’s [4, 5]; the most complete relativistic synthesis is Singh’s [9]. The contributions here are the coefficient-free *bridge* framing (3) and the explicit AeST realization with its four verified checks. An independent proposal of an evolving BTFR normalization—equivalently an evolving a_0 —from a different theory appeared as Marongwe & Kauffman [17]; it adopts the thermal coefficient $a_0 = cH_0/2\pi$ and a steep $e^{-4H_0 t}$ evolution, and is distinguishable from the present $\sqrt{\rho_{\text{DE}}}$ form by both the coefficient ($\sim 11\%$ in a_0) and the evolution steepness (the high- z disc offset magnitude).
2. **The coefficient is data-selected, not derived.** $32\pi = 4 \times 8\pi$: the 8π (Einstein coupling) is forced, leaving the prefactor $4 = (1/2)^2$ of $a_0 = \frac{c}{2}\sqrt{G\rho_{\text{DE}}}$. Writing $a_0 = cH_\Lambda/Z$ with $H_\Lambda = c\sqrt{\Lambda/3}$, the measured a_0 (within the interpolation-function systematic) constrains $Z \in [4.2, 6.0]$, which *excludes* the naive values ($Z = 1$ horizon, $Z = 2$ surface gravity, $Z = \sqrt{8\pi/3} = 2.9$ without the $\frac{1}{2}$) and *includes* $\sqrt{32\pi/3} = 5.79$. Only the $\sim 8\%$ between $\sqrt{32\pi/3}$ and the thermal 2π is genuinely open, and it cancels in the bridge (3). So 32π is well-motivated and data-selected—not free numerology—merely not pinned below $\sim 8\%$.
3. **The covariant coupling is inserted.** $C(Q)$ is consistent and healthy but put in by hand; no symmetry selects it.
4. **Contingent and modest.** The distinctive declining branch is contingent on DESI’s dynamical-DE preference ($\sim 3\text{--}4\sigma$, contested). Under a true constant Λ the prediction is simply constant a_0 .

5. **A theory of the dark universe, not of everything.** The Standard Model is input; gravity is taken to be emergent; the value of Λ (the cosmological-constant problem) and a UV completion of MOND’s strong coupling remain open—as they do for all competitors.

8 Conclusion

If the MOND scale is the cosmological constant, then galaxy dynamics and dark-energy evolution are two views of one quantity, related by the coefficient-free bridge $a_0(z) = a_0(0)\sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)}$. The relation is realized in a concrete, ghost-free, CMB- and GW-safe covariant theory; its distinctive content is decided by a single ~ 30 – 80 -galaxy measurement at $z \sim 3$; and its honest status is a unification of the dark sector that survives present consistency tests while leaving its coefficient, its UV completion, and the Standard Model untouched. We advocate stating it as exactly that.

Data and code availability

The verification scripts (symbolic ghost/CMB/ c_{GW} /universality checks, the bridge and forecast calculations) are available at <https://github.com/carlzimmerman/zimmerman-formula>.

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